

CONTACT

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San Francisco Bay Area

TECHNICAL SKILLS

Spark Hive Hadoop Databricks
Delta Lake Snowflake Redshift dbt
Airflow Dataswarm AWS Azure
GCP Terraform Kubernetes Python
SQL Scala LLM/RAG Pipelines
Claude Code AI Agents Prompt Eng.
LangChain GitHub Copilot MLflow
Diff. Privacy Unity Catalog Data Mesh

EDUCATION

Data Engineering Cert.

UC Santa Cruz, 2013–14

B.S. Electrical Engineering

Sardar Patel College of Eng.

Diploma, Electronics & Comms

Technical Board, Tamil Nadu

PUBLICATIONS

- Cloud Computing
- All About Big Data
- Future Trends in BI

AI & THOUGHT LEADERSHIP

- 160+ articles on paddyspeaks.com
- Claude Code skill development
- Vibe coding methodology
- AI agent architecture writing

OPEN-SOURCE & COMMUNITY

- Interview Studio — free Data & AI Engineering prep
- 22 data-model deep-dives + 1,500+ questions
- In-browser SQL / Python (WASM), no backend

EARLIER CAREER

Data Engineer Consultant

LinkedIn

Dec 2017 – Apr 2018

GDPR compliance — PII encryption into Dali storage (Hive, Python, Pig)

Lead Data Engineer

Meta

Sep 2015 – Sep 2017

80+ Dataswarm pipelines for petabyte-scale ads; Cross Device Insights, Outcomes Datamart + Norms DB

Data Engineer

GREE International

2014 – 2015

PII masking, Vertica → Redshift, 1,000+ table optimization

Data Architect

Chegg Inc.

Oct 2013 – Apr 2014

Technical Director

Model N

2011 – 2013

Life Sciences BI, cloud migration, ETL modularization

Architect

Paddy Iyer

DATA ENGINEERING LEADER · STRATEGIC DATA ARCHITECT · AI & PRIVACY ENGINEERING · AI INNOVATOR

15+ years driving enterprise data transformation for global tech leaders across cloud/SaaS, fintech, gaming, and consumer platforms. Expert in cloud-native architectures, real-time analytics, privacy-first pipelines, and AI-augmented data operations. Proven builder of high-trust, high-performance data teams at petabyte scale. Published writer and thought leader on data architecture, AI agents, and the intersection of ancient wisdom with modern technology.

Modern Data Architecture Privacy-First Design Consent Architecture AI Agents & LLM Pipelines Prompt Engineering Data Governance (C1–C5) Petabyte-Scale Systems Global Team Leadership Data Storytelling

FEATURED EXPERIENCE

Senior Consultant — Privacy & Consent Infrastructure Apr 2024 – Present

Meta | Safe Ads Program

Led the data engineering execution of Meta's Safe Ads Program — a company-critical initiative transitioning advertising infrastructure from broad data access to consent-first architecture across four privacy domains (AAP, Consent Revocation, BYOU, ICOU)

Cross-Namespace Access Control Discovery

- Identified a systemic blind spot in Meta's data deployment stack: when privacy migrations swapped indirect table references to direct cross-namespace sources, pipeline runtime identities crossed policy-zone boundaries for the first time — and no existing pre-deployment check (schema validation, compile tests, or user-identity test runs) could detect the violation
- Recognized that test environments implicitly used engineer credentials with broader access, masking the fact that production pipeline identities lacked cross-zone privileges — a class of failure invisible to all existing CI/CD gates
- Designed and implemented a boundary-guard probe that evaluates cross-namespace policy-zone access at diff time, surfacing violations as hard gates before deployment — preventing silent production outages across 100+ migration diffs
- Established the operational protocol: all cross-namespace table swaps now require explicit privilege verification for the pipeline's runtime identity before landing — a process that did not exist prior to this discovery

AI-Augmented Remediation at Scale

- Co-designed with SWEs a custom AI coding skill (structured playbook) encoding the complete remediation methodology for all four privacy domains — enabling any engineer to execute complex multi-step pipeline migrations with AI assistance from day 2 (vs. ~2 weeks prior ramp-up)
- Cut covers end-to-end: table transform generation, consent filter pipeline creation, signal table dependency wiring, multi-producer alignment, lint/test automation, and cross-team dependency resolution
- Cut per-asset remediation time by 3–5× and enabled a single engineer to sustain 800+ landed diffs over the engagement — a throughput level typically requiring 3–4 engineers without AI augmentation
- Collaborated with software engineers to continuously improve the AI implementation: expanded coverage from AAP-only to CR, BYOU, and ICOU domains; added cross-namespace boundary detection; eliminated fragile metadata tracking patterns
- Applied prompt engineering and LLM-assisted code analysis to reduce documentation overhead by ~70%, freeing engineering time for higher-judgment work (policy-zone gap analysis, multi-producer negotiations)

Privacy Remediation Pipeline Architecture

- Architected an 8-stage remediation pipeline (classification → lineage → topology sort → pipeline dev → diff → deploy → compliance → downstream migration) serving as the execution framework for 700+ data asset remediations
- Achieved 100% remediation of all high-risk assets across AAP and Consent Revocation on schedule — directly unblocking Meta's Safe Ads regulatory launch
- Built consent filtering pipelines enforcing user-level opt-out at the data layer, signal table architectures for auditable dependency management, and differential privacy operators preserving analytical utility under consent constraints

CallidusCloud
2005 – 2011
Incentive comp analytics, BusinessObjects
XI

DW Architect
Hewlett Packard
2001 – 2005
Enterprise DW + 8 datamarts; ETL 18 → 3
hrs; 40–50% sales lift via clickstream

- Closed compliance gaps across cross-functional pipelines involving 3+ producer teams — required navigating ambiguous ownership, negotiating schema changes, and resolving conflicting team priorities
- Reduced average asset remediation cycle time from weeks to days; cut time-to-first-deploy for new pipelines by ~50% through documented, repeatable validation workflows

EARLIER CAREER

Lead Data Architect — Partner Data Engineering *Jan 2022 – Jan 2024* VMware

Strategic architect for VMware's Partner Data Platform — unifying fragmented partner data into a single governed analytics layer

- Designed and built the Partner Data Platform from the ground up, unifying partner data from 10+ disparate sources into a single governed analytics layer — eliminating silos that had blocked cross-functional decision-making for years
- Implemented VMware's first structured data governance for partner data (classification, lineage, policy enforcement) plus a self-service data catalog — cutting tribal-knowledge reliance by 60% and ad-hoc request volume by ~40%, and achieving audit-ready status
- Pioneered an AI copilot for retrieval optimization, coding, and governance — one of VMware's earliest LLM-assisted data engineering deployments, reducing routine query development time by ~30%
- Achieved 40%+ Spark processing improvements via broadcast joins, caching, and skew management — bringing SLA-critical reports inside their delivery windows for the first time
- Engaged director/VP-level stakeholders to define a unified data vision, securing budget and headcount that grew the team from 3 to 8 engineers during the engagement

Senior Consultant — Ads, Commerce & Privacy *Jul 2019 – Dec 2021* Meta

Led high-impact projects across Meta's Ads, Commerce, and Privacy teams, architecting secure, scalable, and compliant data platforms

- Led privacy remediation across 1,200+ SMB 2.0, Customer Journey, and BPO tables — resolving critical compliance-risk design flaws and establishing remediation patterns later adopted at scale in the Safe Ads program
- Spearheaded migration of 1,000+ Hive pipelines to Spark, automating ~70% of conversion tasks — compressing a projected 12-month migration to under 6 months
- Owned the SMB funnel redesign, improving query performance 3x and scaling the pipeline from thousands to tens of thousands of small-business advertisers without infrastructure changes
- Enhanced Hive anonymization processes with privacy engineers, enabling ads and commerce data products to pass privacy audits without manual intervention — cutting audit preparation effort by ~60%